MINISTRY OF
ELECTRONICS &
INFORMATION TECHNOLOGY
GOVERNMENT OF INDIA

सत्यमेव जयते

CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING (C-DAC)

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Idea Forge

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Problem Understanding

The Last-Mile Civic Service Problem

- Citizens spend hours visiting multiple offices for basic services (electricity, gas, water, municipal)
- No unified system → fragmented, inefficient experience

Existing kiosks

- Single-service only
- English-only
- Not usable by elderly / low-literacy users

Complaint systems lack

- Transparency
- Tracking
- Accountability

Core Gap

- No offline-capable,
- Inclusive,
- Unified civic platform at the grassroots level

- No tamper-proof audit system → risk of corruption & disputes

Proposed Solution

SUVIDHA Connect – Civic OS for Bharat

- A Smart, Unified Self-Service Kiosk Platform

What We Built:

Unified Pf

- Electricity
- Gas
- Water
- Municipal

Secure & Seamless:

- OTP / QR login
- Real-time bill payment
- Complaint + tracking system

Smart Capabilities:

- AI Chat Assistant (SUVIDHA Sahayak)
- Voice Commands (Hindi + English)
- Text-to-Speech (Accessibility-first)

Technology:

- Progressive Web App (PWA)
- Works offline + online

Impact:

- Reduce queues, increase transparency, enable self-service governance

System Architecture

Scalable & Secure Architecture

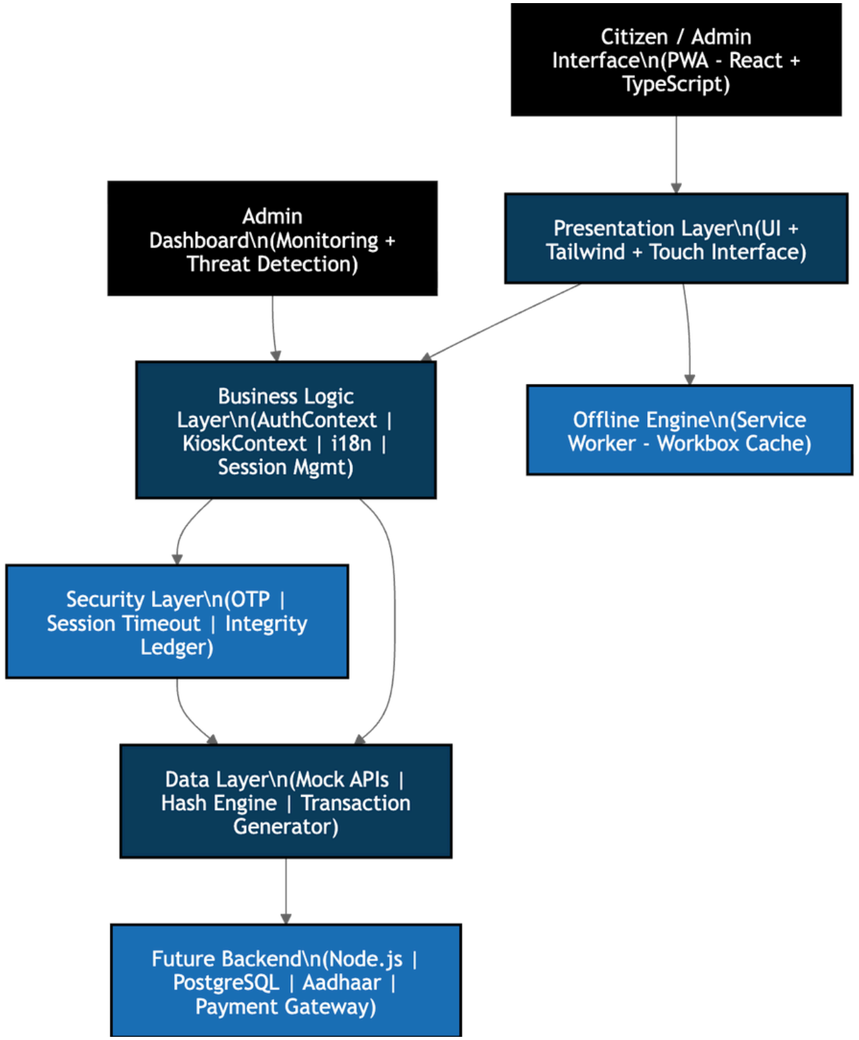
Layered Architecture

Presentation Layer

Business Logic Layer

Data Layer

Security Built-In	Future Ready
• 3-min session timeout (activity-based)	REST APIs + PostgreSQL
• OTP authentication	Aadhaar eKYC + Payment G
• Integrity Ledger (hash chaining)	Real-Time Updates



UI/UX (Kiosk Focused Security, Scalability & Compliance)

Designed for Bharat, Not Just Users

Kiosk-First UX

- Large touch buttons (48px+)
- Virtual keyboard (no hardware dependency)
- Simple, guided navigation

Accessibility

- Voice navigation (Hindi + English)
- Text-to-Speech
- High contrast mode
- Senior-friendly UI

Security by Design

- Session auto-logout (3 min)
- Aadhaar masking
- CAPTCHA + input sanitization
- Full audit logging per kiosk

Scalability & Complaince

- Modular architecture
- Multi-kiosk deployment with unique IDs
- Offline-first & DPDP Compliance

Innovation & Future Scope

Innovation that Drives Adoption

Integrity Ledger	Voice AI System	Emergency Detection	Gamification	Offline PWA
Blockchain-style hash chain for every transaction	Speak → Understand → Respond → Read aloud	Auto escalation for keywords like "fire", "ambulance"	Rewards for civic participation	Works even without internet

Future Scope

- Aadhaar eKYC integration
- UPI / BBPS payments
- Hyperledger blockchain backend
- 10+ regional languages
- IoT smart meter integration
- WhatsApp/SMS updates

Conclusion

THANK

YOU